CORRECTION TO THE LEVEL97 CERTIFICATES IN THE S(15) AND S(16) PROOFS

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The earlier code used

alpha=sigma^6-sigma^4-sigma^2+1,

where sigma is the primitive diamond at97. This polynomial equals (sigma^2-1)*(sigma^4-1) and does not vanish at a primitive12th root. It therefore did not supply the finite-rational-image statement asserted in the proof. The narrative factorization involving Phi_12 did not match those code coefficients.

The corrected computation uses alpha=sigma^12-1 and, on the three closed orbits of size24 in degrees10/14, (sigma-1)*Phi12(sigma). Both annihilate the
trivial character and every strange character at97, since the only strange character order is12 (Derickx--Stoll, Table2). Consequently each operator has finite image on J_1(97)(Q), and composing with the good-prime torsion annihilator at3 kills J_1(97)(Q).

All359 ordinary residue orbits through degree17 have been recomputed with these valid operators. Every resulting principal-divisor obstruction has full rank, independently checked in Sage. Restricting to degree<=15 gives89 matrices; restricting to degree<=16 gives176. Their modular-form basis and source metadata agree exactly with the earlier models. The earlier stored level97 matrices, code and proof PDFs have been replaced accordingly, and both complete coverage checks pass again with no unresolved primes.

The results S(15)=Primes(43) and S(16)=Primes(37) are unchanged. The computations through S(14) used a valid level97 polynomial and are not affected by this error. The separate supersingular level97 arguments in degree16 are also unaffected.

Authoritative files:

- work/degree17/full97/sieve.json
- work/degree17/full97/matrices.json.gz
- work/degree17/independent_rank_checks.json
- work/degree17/level97_correction_verified.json
- work/degree15/verify_coverage_after97_correction.log
- work/degree16/verify_coverage_after97_correction.log